AP PHYSICS 1

Unit 1 • Worksheet 1.1A

Worksheet 1.1A • Scalars and Vectors in One Dimension

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

Which of the following quantities is a vector?

(A) Speed
(B) Distance
(C) Displacement
(D) Time

Question 2

Which of the following quantities is a scalar?

(A) Velocity
(B) Acceleration
(C) Force
(D) Mass

Question 3

An object moves along the x-axis from $x = +3\,\text{m}$ to $x = -5\,\text{m}$. What is its displacement?

(A) $+8\,\text{m}$
(B) $-8\,\text{m}$
(C) $-2\,\text{m}$
(D) $+2\,\text{m}$

Question 4

A person walks 30 m east and then 50 m west. What are the distance traveled and the displacement?

(A) 80 m; 20 m west
(B) 80 m; 20 m east
(C) 20 m; 80 m west
(D) 20 m west; 80 m

Question 5

East is positive. Which velocity has the greatest speed?

(A) $+6\,\text{m/s}$
(B) $-9\,\text{m/s}$
(C) $+8\,\text{m/s}$
(D) $-4\,\text{m/s}$

Question 6

A ball is thrown straight up and caught at the same height. What is its displacement for the whole trip?

(A) Twice the maximum height
(B) Zero
(C) The maximum height
(D) Negative

Question 7

Up is positive. A stone falls 12 m from the top of a cliff to the ground. What is its displacement?

(A) $+12\,\text{m}$
(B) $0\,\text{m}$
(C) $-12\,\text{m}$
(D) $24\,\text{m}$

Question 8

Two forces act on a box along the same line: 40 N to the right and 25 N to the left. What is the net force?

(A) 65 N to the right
(B) 15 N to the left
(C) 15 N to the right
(D) 65 N to the left

Question 9 (Free response)

A car on a straight road starts at $x = 0$, drives to $x = +120\,\text{m}$, then back to $x = +40\,\text{m}$, then forward to $x = +70\,\text{m}$.

(a) Find the total distance traveled.
(b) Find the displacement.
(c) Can the magnitude of a displacement ever be larger than the distance traveled? Explain.